

AI Powered Debt Relief & Financial Recovery Platform

Project Description:

The AI Powered Debt Relief & Financial Recovery Platform is an intelligent AI-powered web application developed to simplify and automate the debt management and financial recovery process for borrowers. The platform enables users to manage loan details, analyse financial health, generate AI-powered negotiation strategies, and receive adaptive settlement recommendations through a secure and modern digital environment.

Built using React, JS, Fast API, Python, SQLite, SQL Alchemy ORM, and Google Gemini API, the system provides a responsive, scalable, and borrower-friendly interface with intelligent financial processing capabilities and real-time AI-assisted decision support.

Scenarios

Scenario 1: AI-Powered Settlement Recommendation

A borrower adds loan details including outstanding amount, EMI, overdue duration, and monthly income. The system analyses the financial profile and generates personalized settlement recommendations along with debt stress analysis and financial health insights.

Scenario 2: Intelligent Negotiation Letter Generation

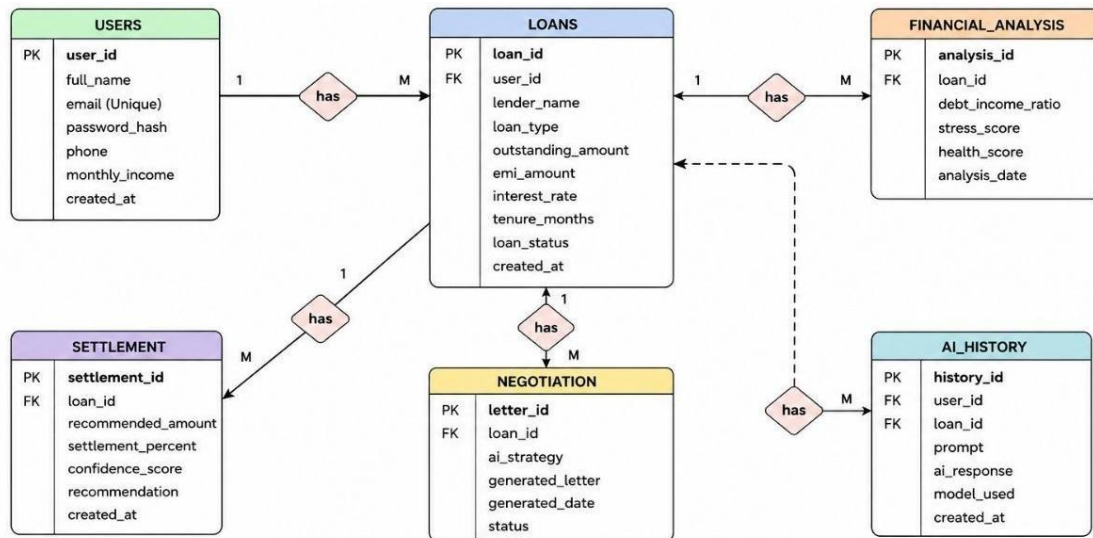
A borrower selects a loan account and requests a negotiation strategy. The platform uses Google Gemini AI to generate professional lender-specific negotiation emails and settlement request letters based on the borrower's financial condition and repayment capability.

Scenario 3: Financial Health Tracking & Loan Analysis

A borrower accesses the dashboard to monitor financial health metrics such as monthly surplus, EMI ratio, debt stress level, and settlement percentages. The platform also maintains AI-generated negotiation history and provides borrower guidance for improving debt management decisions.

Technical Architecture:

ER DIAGRAM - AI POWERED DEBT RELIEF & FINANCIAL RECOVERY PLATFORM



Description:

The AI Powered Debt Relief & Financial Recovery Platform is designed to help borrowers manage their debts, analyse financial health, and generate personalized settlement strategies using artificial intelligence. Users register on the platform and add their loan details, including lender information, outstanding amount, EMI, interest rate, and tenure. The system performs financial analysis on each loan to calculate key metrics such as debt-to-income ratio, stress score, and financial health score. Based on this analysis, the platform uses AI models to predict optimal settlement amounts and recommend negotiation strategies. It also generates professional negotiation letters that borrowers can use to communicate with lenders. All AI interactions, prompts, and responses are securely stored in the AI History table for transparency, tracking, and future reference. This relational database structure ensures data consistency, scalability, and efficient management of users, loans, financial insights, and AI-driven recommendations.

Pre-requisites:

Python (Programming Language)

Fast API

SQLite

Generative AI

Responsible AI

LLMs

Prompt Engineering

Gemini AI

API Integration

Cloud Computing

Project Workflow:

Milestone 1: Model Selection and Architecture

- **Activity 1.1 Requirement Analysis**
The project requirements were analysed to identify the system objectives, user needs, and technical constraints before development.
- **Activity 1.2 Technology Selection**
React.js, Fast API, SQLite, SQL Alchemy, and Google Gemini AI were selected to build a scalable and AI-powered application.
- **Activity 1.3 System Architecture Design**
A three-tier architecture was designed to enable secure communication between the frontend, backend, and database.

Milestone 2: Core Functionalities Development

- **Activity 2.1 User Authentication and Loan Management**
A secure authentication system was implemented, allowing users to register, log in, and manage their loan information efficiently.
- **Activity 2.2 AI Financial Analysis and Settlement Recommendation** Google Gemini AI analyses financial data and provides personalized settlement recommendations along with negotiation strategies.
- **Activity 2.3 Negotiation Letter Generator**
Borrower details, current balance, proposed settlement

Milestone 3: App.py Development

- **Activity 3.1 Fast API Configuration and API Development**
The [app.py](#) file initializes the Fast API application and defines REST API endpoints for all system functionalities.
- **Activity 3.2 Database and Gemini AI Integration**
The backend connects with the SQLite database to store user data and integrates Gemini AI to generate intelligent financial insights.

Milestone 4: Frontend Development

- **Activity 4.1 User Interface Design and API Integration**
A responsive React.js interface was developed and integrated with backend APIs to provide a seamless user experience.

Milestone 5: Deployment

- **Activity 5.1 Environment Configuration**
The deployment environment was configured with the required software dependencies and environment variables.
- **Activity 5.3 Security Configuration**
Authentication mechanisms and secure API communication were implemented to protect user information.
- **Activity 5.4 System Testing**
The deployed application was tested to verify functionality, performance, and overall system reliability.

Milestone 6: Conclusion

- **Activity 6.1 Conclusion**
The AI Powered Debt Relief & Financial Recovery Platform successfully combines modern web technologies with artificial intelligence to simplify debt management. The developed system provides secure, scalable, and intelligent financial assistance while offering a strong foundation for future enhancements.

Milestone 1: Model Selection and Architecture

Activity 1.1 Requirement Analysis

The project requirements were analysed to understand the financial challenges faced by borrowers and to design an AI-assisted debt management system. The analysis helped identify the essential features required for effective financial recovery.

- Identified user requirements for debt management.
- Analysed existing debt settlement challenges.
- Defined project objectives and system scope.
- Planned AI integration for financial recommendations.

Activity 1.2 Technology Selection

Layer	Technology	Purpose
Frontend	React, Vite, CSS	Responsive form-driven borrower interface.
Backend	Python, Fast API	REST APIs, validation and business logic.
Data	SQLite, SQL Model	Persistence for users, loans, predictions and letter history.
Authentication	JWT and password hashing	Protected user sessions.
Deployment	Docker and Docker Compose	Repeatable local full-stack execution.

The technology stack was selected to ensure high performance, scalability, and ease of development. Modern web technologies and AI services were chosen to build an efficient financial recovery platform.

- React.js for responsive frontend development.
- Fast API for high-performance backend APIs.
- SQLite with SQL Alchemy ORM for database management.
- Google Gemini AI for intelligent financial analysis.

Activity 1.3 System Architecture Design

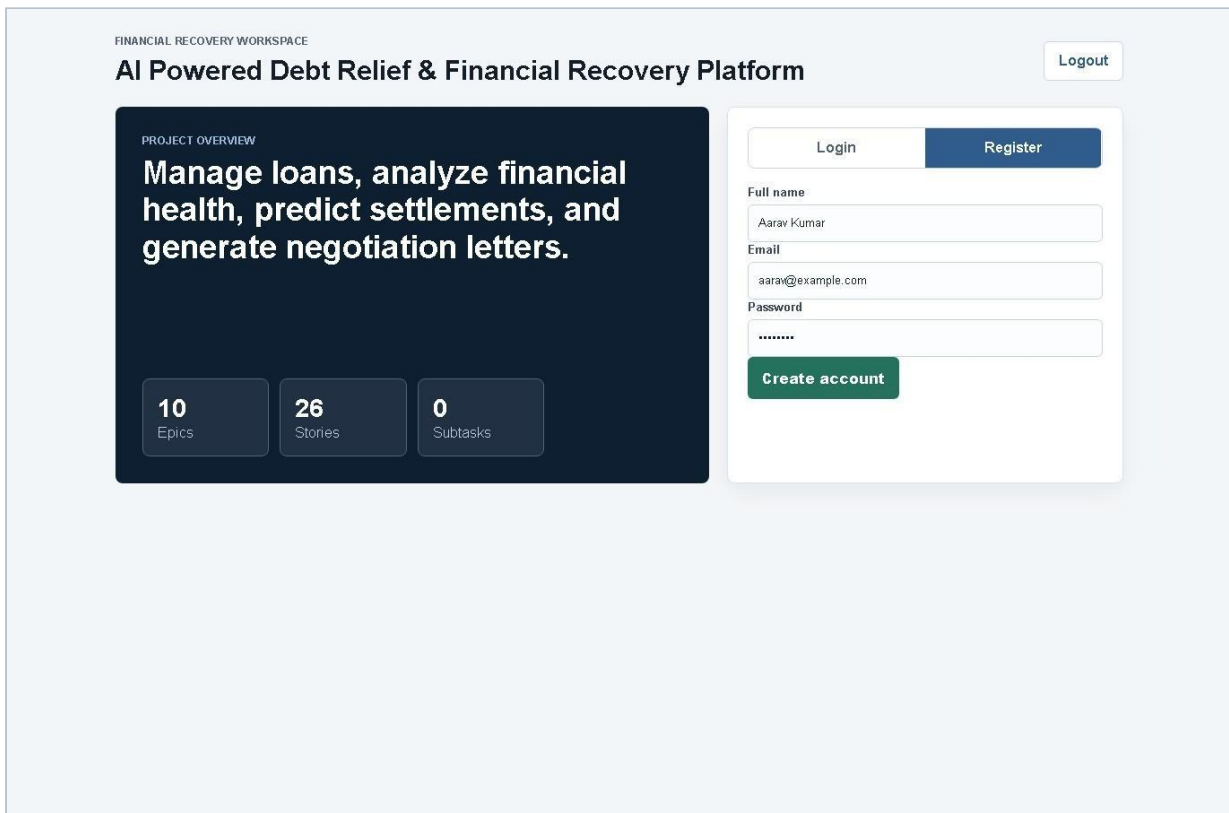
A three-tier architecture was implemented to separate the presentation, business logic, and database layers. This design improves maintainability, security, and system performance.

- Designed frontend, backend, and database layers.
- Established REST API communication.
- Integrated AI services with the backend.
- Ensured modular and scalable architecture.

Milestone 2: Core Functionalities Development

Activity 2.1 User Authentication and Loan Management

Users can create an account or log in using email and password. A successful response stores a JWT token, allowing access to the protected workspace.



The image shows a web application interface for a "FINANCIAL RECOVERY WORKSPACE". The main heading is "AI Powered Debt Relief & Financial Recovery Platform". In the top right corner, there is a "Logout" button. On the left, a dark blue panel titled "PROJECT OVERVIEW" contains the text "Manage loans, analyze financial health, predict settlements, and generate negotiation letters." Below this text are three boxes showing statistics: "10 Epics", "26 Stories", and "0 Subtasks". On the right, there is a registration form with two tabs: "Login" and "Register". The "Register" tab is active. The form includes input fields for "Full name" (with the example "Aarav Kumar"), "Email" (with the example "aarav@example.com"), and "Password" (with masked characters "*****"). A green "Create account" button is at the bottom of the form.

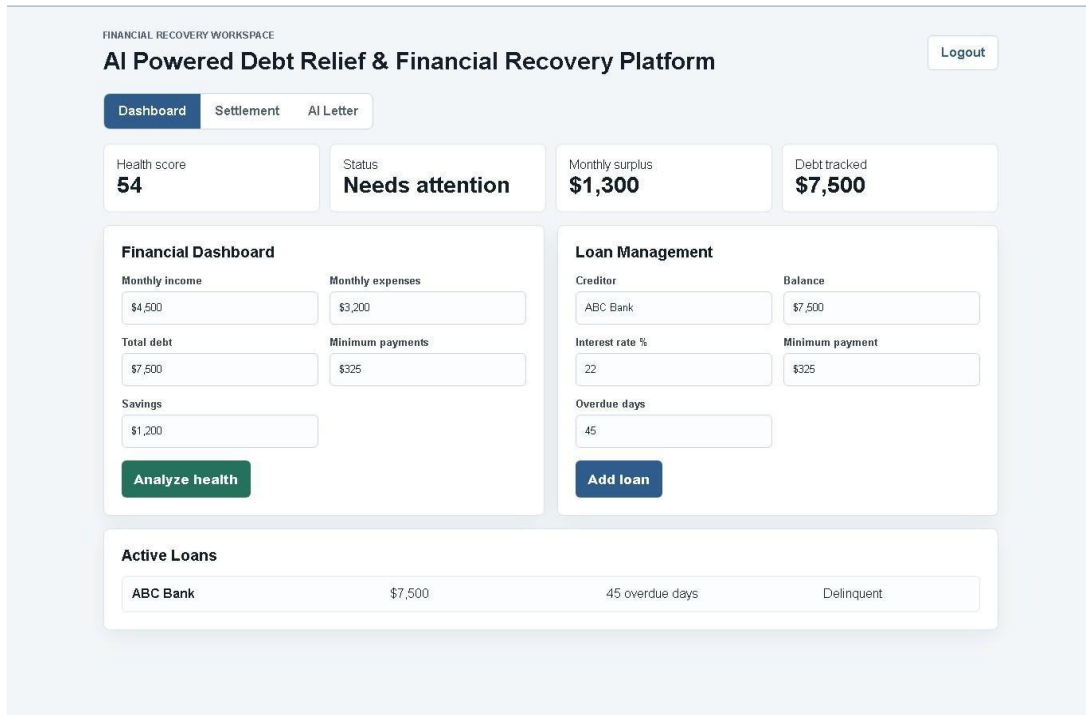
Figure 1. Registration view with the project overview panel and account form.

The application provides a secure authentication system and enables users to manage multiple loans efficiently. All loan information is securely stored and retrieved for financial analysis.

- User registration and login.
- Secure password authentication.
- Add, edit, and delete loan records.
- Store lender and EMI information.

Activity 2.2 AI Financial Analysis and Settlement Recommendation

The dashboard collects income, expenses, total debt, minimum payments and savings. It displays the financial-health score, status, monthly surplus and tracked debt. Loan management records creditor, balance, interest rate, minimum payment and overdue days.



FINANCIAL RECOVERY WORKSPACE

AI Powered Debt Relief & Financial Recovery Platform

Logout

Dashboard Settlement AI Letter

Health score
54

Status
Needs attention

Monthly surplus
\$1,300

Debt tracked
\$7,500

Financial Dashboard

Monthly income
\$4,500

Monthly expenses
\$3,200

Total debt
\$7,500

Minimum payments
\$325

Savings
\$1,200

Analyze health

Loan Management

Creditor
ABC Bank

Balance
\$7,500

Interest rate %
22

Minimum payment
\$325

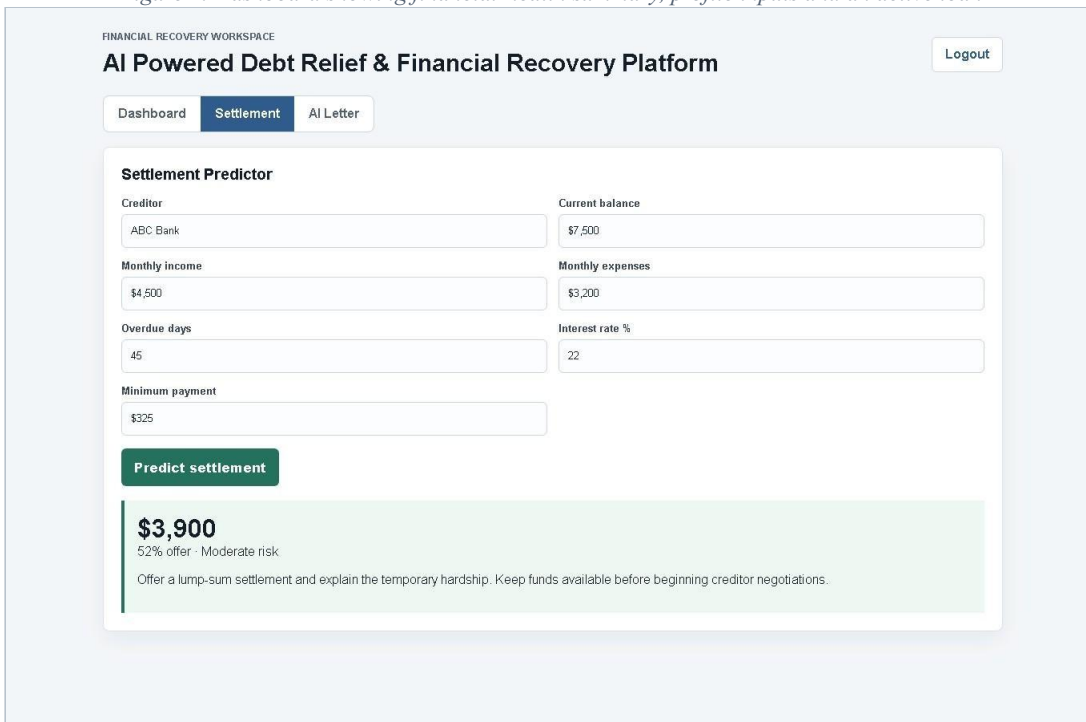
Overdue days
45

Add loan

Active Loans

ABC Bank	\$7,500	45 overdue days	Delinquent
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Figure 2. Dashboard showing financial-health summary, profile inputs and an active loan



FINANCIAL RECOVERY WORKSPACE

AI Powered Debt Relief & Financial Recovery Platform

Logout

Dashboard Settlement AI Letter

Settlement Predictor

Creditor
ABC Bank

Current balance
\$7,500

Monthly income
\$4,500

Monthly expenses
\$3,200

Overdue days
45

Interest rate %
22

Minimum payment
\$325

Predict settlement

\$3,900
52% offer - Moderate risk
Offer a lump-sum settlement and explain the temporary hardship. Keep funds available before beginning creditor negotiations.

Figure 3. Settlement prediction with a representative suggested offer and strategy.

The system analyses the user's financial data and generates personalized debt settlement recommendations using Google Gemini AI. It also produces professional negotiation letters for communication with lenders.

- Analyses financial health.
- Calculate debt burden.
- Generate AI-based settlement strategies.
- Create professional negotiation letters.

Activity 2.3 Negotiation Letter Generator

FINANCIAL RECOVERY WORKSPACE

Logout

Dashboard
Settlement
AI Letter

AI Negotiation Letter Generator

Your name Aarav Kumar	Creditor ABC Bank
Current balance \$7,500	Settlement offer \$3,900
Monthly income \$4,500	Hardship reason temporary reduction in income

Generate letter

Dear ABC Bank Settlement Team,

I am writing regarding my current balance of \$7,500. Due to a temporary reduction in income, I am unable to maintain the existing payment arrangement.

I can offer a one-time settlement payment of \$3,900 in full resolution of this account. Please confirm the terms in writing before payment.

Sincerely,
Aarav Kumar

Figure 4. AI letter form and generated hardship-based settlement request.

The letter generator converts borrower details, current balance, proposed settlement and hardship reason into a respectful lender-ready draft. The user can review the text before using it.

Milestone 3: App.py Development

Activity 3.1 Fast API Configuration and API Development

The app.py file acts as the central backend module by configuring the Fast API application and defining all API endpoints required for the project.

- Initialize Fast API server.
- Configure REST API routes.
- Handle HTTP requests and responses.
- Implement exception handling.

Activity 3.2 Database and Gemini AI Integration

The backend communicates with the SQLite database to manage user information and integrates with Google Gemini AI to generate intelligent financial recommendations.

- Database connection using SQL Alchemy.
- CRUD operations for loan data.
- Gemini API integration.
- AI response processing.

Milestone 4: Frontend Development

Activity 4.1 User Interface Design and API Integration

React frontend → Fast API REST API → financial scoring, settlement and letter utilities → SQL Model / SQLite. The frontend calls endpoints for authentication, loans, financial-health analysis, settlement prediction and letter generation. The backend validates inputs, applies the project's transparent rules, and persists relevant records.

A responsive and interactive frontend was developed using React.js. The frontend communicates with the backend through REST APIs to display real-time financial information.

- Responsive dashboard design.
- Login and registration pages.
- Loan management interface.
- API integration with Fast API.

Milestone 5: Deployment

Activity 5.1 Environment Configuration

The deployment environment was configured by installing required dependencies and securely managing application settings.

- Configure project dependencies.
- Set environment variables.
- Configure database settings.
- Add Gemini API credentials.

Activity 5.2 Backend and Frontend Deployment

The backend and frontend applications were deployed to make the platform accessible over the internet.

- Deploy Fast API backend.
- Host React.js frontend.
- Verify API communication.
- Test deployed application.

Activity 5.3 Security Configuration

Security measures were implemented to protect user data and prevent unauthorized access to the application.

- JWT authentication.
- Password encryption.
- Secure API communication.
- Input validation.

Activity 5.4 System Testing

The deployed application was tested thoroughly to verify functionality, performance, and reliability before final deployment.

- Functional testing.
- API testing.
- User interface testing.
- Performance verification.

Milestone 6: Conclusion

Activity 6.1 Conclusion

The AI Powered Debt Relief & Financial Recovery Platform successfully combines Artificial Intelligence with modern web technologies to provide intelligent debt management solutions. The platform enables users to analyse their financial condition, receive AI-powered settlement recommendations, and generate professional negotiation letters. Its scalable architecture and secure implementation make it a reliable solution for future financial assistance applications.

- AI-powered financial guidance.
- Secure and scalable architecture.
- Automated debt settlement recommendations.
- Improved user experience and financial decision-making.